CRNS Site Characterization Tool

A Physics-Based Pre-Installation Assessment Pipeline for Cosmic-Ray Neutron Sensing Stations

M. Barbieri

March 22, 2026

Technical Note — internal document.
Source code:
https://github.com/barbierimauro/site_characterization

Contents

Abstract **3**

1 Introduction **4**
1.1 Cosmic-Ray Neutron Sensing . . . 4
1.2 The Need for Pre-Installation Site Characterisation . . . 4
1.3 Existing Tools and Gaps . . . 4
1.4 Scope of This Document . . . 5

2 Theoretical Background **6**
2.1 Neutron Footprint Geometry . . . 6
2.2 The Desilets Inversion Formula . . . 6
2.3 Topographic Correction κ_{topo} . . . 7
2.4 Muon Field-of-View Correction κ_{muon} . . . 7
2.5 Land-Use/Land-Cover Correction κ_{lulc} . . . 8
2.6 Site Flux Estimates and N_0 . . . 8

3 Data Sources **9**
3.1 Digital Elevation Model . . . 11
3.2 Land-Use / Land-Cover . . . 11
3.3 Soil Properties . . . 11
3.4 Atmospheric Data . . . 11
3.5 Soil Moisture — ERA5-Land . . . 12
3.6 Vegetation (LAI) . . . 12
3.7 Geology . . . 12

4 Computational Methods **13**
4.1 Pipeline Overview . . . 13
4.2 Terrain Processing . . . 13
4.3 Topographic Correction κ_{topo} . . . 14
4.4 LULC Correction and Supplementary Hydrogen . . . 14
4.5 Supplementary Correction Factors . . . 14
4.6 Optimal Soil Sampling Plan . . . 15
4.7 Soil Moisture Downscaling and Data Fusion . . . 15
4.8 Geological Characterisation . . . 16

5 Worked Example: Italian Alps Test Site **18**
5.1 Terrain and Topographic Correction . . . 18
5.2 Muon Field-of-View Correction . . . 18
5.3 Land-Use / Land-Cover Correction . . . 18
5.4 Total Correction and Reference Count . . . 19
5.5 Soil Properties and Supplementary Hydrogen Inventory . . . 19
5.6 Atmospheric and Vegetation Seasonal Climatology . . . 19
5.7 Optimal Sampling Plan . . . 20

6 Limitations and Uncertainties **21**
6.1 DEM Resolution . . . 21
6.2 LULC Classification . . . 21
6.3 Soil Properties . . . 21
6.4 Lattice Water Estimate . . . 21
6.5 Atmospheric Corrections . . . 21
6.6 Geological Data . . . 22

6.7 Geomagnetic Rigidity 22
6.8 Combined Uncertainty on N_0 22

7 Conclusions **22**

A List of Variables and Constants **25**

B Pipeline Pseudocode **27**

Abstract

Cosmic-Ray Neutron Sensing (CRNS) is a field technique for non-invasive, area-averaged soil moisture measurement at the hectometre scale. Before deploying a CRNS station, a thorough characterisation of the site is necessary to (i) quantify the geometric and geophysical corrections that relate measured neutron counts to soil moisture, (ii) estimate the expected signal dynamic range, and (iii) design the optimal soil sampling campaign for calibration.

This technical note describes a fully integrated Python pipeline that automates CRNS site characterisation from publicly available global datasets. The core physical outputs are three multiplicative correction factors: the topographic correction κ_{topo}, computed by a three-dimensional ray-casting algorithm over a 30-m DEM; the muon field-of-view correction κ_{muon}, derived analytically from the horizon angle profile; and the land-use/land-cover correction κ_{lulc}, based on ESA WorldCover and OpenStreetMap. Their product $\kappa_{\text{tot}} = \kappa_{\text{topo}}\,\kappa_{\text{muon}}\,\kappa_{\text{lulc}}$ directly enters the calibration of the reference count N_0 through the Desilets (2010) equation, extended here to account for clay-mineral lattice water l_w (Köhli et al., 2021) while preserving full analytical invertibility.

Supplementary characterisation modules provide: seasonal atmospheric water-vapour and snow-water-equivalent climatologies from ERA5; above-ground biomass hydrogen estimates from MODIS LAI; global lithological information from the Macrostrat API; an optimally weighted soil sampling plan; and a Topographic Wetness Index.

The pipeline is validated on a test site in the Italian Alps, demonstrating how each correction factor and supplementary quantity can inform installation decisions and calibration planning.

1 Introduction

1.1 Cosmic-Ray Neutron Sensing

The Cosmic-Ray Neutron Sensing (CRNS) method exploits the natural flux of epithermal neutrons produced when high-energy cosmic-ray particles interact with the Earth's atmosphere and surface [Zreda et al., 2012]. Secondary neutrons that reach the ground are moderated primarily by hydrogen-bearing substances; liquid soil water is by far the largest and most variable hydrogen pool in most natural environments. A stationary CRNS probe counts epithermal neutrons continuously, and the count rate N is inversely related to the volumetric soil moisture θ_v over an area roughly $\sim$100 m in radius and $\sim$10 cm in depth [Desilets et al., 2010, Köhli et al., 2015].

This footprint is orders of magnitude larger than a conventional point-scale TDR or FDR sensor yet smaller than remote-sensing products, making CRNS uniquely suited to bridge the gap between in-situ hydrology and satellite observations. The technique is now established globally through networks such as COSMOS [Zreda et al., 2012] and TERENO, and is increasingly used for irrigation scheduling, drought monitoring, and land-surface model validation.

1.2 The Need for Pre-Installation Site Characterisation

A CRNS measurement is not simply a count of neutrons: it is the result of many simultaneous physical effects that must be corrected before soil moisture can be retrieved. Some corrections are *operational* — they must be applied to each measurement epoch and require ancillary meteorological data (barometric pressure, atmospheric humidity, incoming cosmic-ray intensity monitor). Others are *site-specific constants* that can be determined once, before installation:

- **Topographic correction** κ_{topo}: departures from a flat reference change the soil volume seen by the probe. A sensor on a ridge sees less soil than expected; one at the bottom of a valley may see more.
- **Muon field-of-view correction** κ_{muon}: high-energy muons (which produce part of the secondary-neutron flux) are blocked by surrounding terrain above the horizon, reducing the effective production rate.
- **Land-use/land-cover correction** κ_{lulc}: water in vegetation, roads, buildings, or standing water within the footprint contributes to neutron moderation and must be accounted for.
- **Soil hydrogen inventory**: clay-mineral lattice water (l_w), soil organic carbon (SOC), and bulk density (ρ_b) set the background hydrogen level against which liquid water is detected.

Without these corrections, N_0 — the calibration constant of the Desilets inversion — is biased, and all subsequent soil moisture estimates carry a systematic error. Pre-installation site characterisation is therefore a prerequisite for accurate, quantitative CRNS measurements.

1.3 Existing Tools and Gaps

Several components of CRNS data processing are available as standalone libraries or web services (e.g. the `crnpy` Python package for operational corrections, the COSMOS data portal). However, no openly available, integrated tool exists that: (i) downloads all relevant datasets automatically from global open sources, (ii) computes all site-specific correction factors with physically consistent methods, (iii) characterises climate, vegetation, soil, geology, and snow cover for a given location, and (iv) produces a ready-to-use calibration curve and sampling plan.

1.4 Scope of This Document

This technical note describes the *CRNS Site Characterization Pipeline* (`crns-sitechar`), a Python 3 tool that addresses the gap above. Section 2 presents the theoretical framework. Section 3 summarises all data sources. Section 4 describes the computational methods. Section 5 shows a worked example. Section 6 discusses limitations and uncertainties. Section 7 concludes.

2 Theoretical Background

2.1 Neutron Footprint Geometry

The neutron footprint of a CRNS probe is characterised by two length scales: the radial sensitivity radius r_{86} and the penetration depth z_{86}. Both depend on the total hydrogen content in the environment, which includes liquid soil water θ_v, clay-mineral lattice water l_w, and soil organic matter.

Footprint radius. Köhli et al. [2015] showed that the radius enclosing 86% of the neutron signal is well approximated by

$$r_{86}(\theta_v, P) = \frac{A_1}{\theta_v + A_2} \cdot \frac{P_0}{P}, \tag{1}$$

with $A_1 = 29.13\,\mathrm{m\,m^3\,m^{-3}}$, $A_2 = 0.0578\,\mathrm{m^3\,m^{-3}}$, reference pressure $P_0 = 1013.25\,\mathrm{hPa}$, and site pressure P. At $\theta_v = 0.20\,\mathrm{m^3\,m^{-3}}$ and sea level, $r_{86} \approx 113\,\mathrm{m}$. The footprint shrinks by roughly 40% from dry ($\theta_v \to 0$) to saturated ($\theta_v \approx 0.45\,\mathrm{m^3\,m^{-3}}$) conditions.

Penetration depth. The depth enclosing 86% of the neutron signal is

$$z_{86}(\theta_v, \rho_b, l_w) = \frac{8.3\,\mathrm{cm}}{\rho_b(0.0564 + \theta_v + l_w)}, \tag{2}$$

where ρ_b is the dry soil bulk density [$\mathrm{g\,cm^{-3}}$], θ_v is liquid volumetric water content [$\mathrm{m^3\,m^{-3}}$], and l_w is the lattice water content [$\mathrm{g\,g^{-1}}$] [Bogena et al., 2013, Köhli et al., 2021]. Including l_w in Eq. (2) is physically necessary because clay-mineral-bound hydrogen moderates neutrons even in air-dry soil, reducing z_{86} regardless of liquid water.

Radial sensitivity weight. The contribution of a soil volume element at radial distance r from the probe decreases exponentially [Köhli et al., 2015]:

$$W(r) = \exp\left(-\frac{r}{r_{86}/3}\right). \tag{3}$$

This weight function is used throughout the pipeline to compute footprint-averaged quantities (TWI, LULC fractions, soil properties).

2.2 The Desilets Inversion Formula

The relationship between neutron count rate N and liquid soil moisture θ_v is described by the Desilets equation [Desilets et al., 2010]:

$$\frac{N}{N_0} = a_0 + \frac{a_1}{\theta_v + l_w + a_2}, \tag{4}$$

with parameters $a_0 = 0.0808$, $a_1 = 0.372$, $a_2 = 0.115$ [Desilets et al., 2010]. The term l_w has been added here (Köhli et al., 2021): it represents the constant offset from lattice water, which is indistinguishable from liquid water by the neutron probe. The formula retains full analytical invertibility:

$$\theta_v = \frac{a_1}{N/N_0 - a_0} - l_w - a_2. \tag{5}$$

This is the key advantage of Eq. (4) over more complex parameterisations [Köhli et al., 2015]: N_0 can be determined unambiguously from a single calibration measurement.

Reference count N_0. N_0 is computed from the theoretically expected epithermal neutron count at the site under "dry" conditions ($\theta_v = 0$, only l_w present):

$$N_0 = \frac{N_{\text{neut,site}}}{a_0 + a_1 \,/\, (lw + a_2)}, \tag{6}$$

where $N_{\text{neut,site}}$ is the absolute expected count rate after altitude, rigidity, and topographic corrections (Section 2.6). With $l_w = 0$, Eq. (6) reduces to the original Desilets expression.

Supplementary hydrogen inventories. Two additional hydrogen pools can bias N_0 estimates but are treated as supplementary information rather than modifications to Eq. (4): (i) soil organic carbon contributes an effective hydrogen equivalent $\theta_{\text{SOC}} \approx 0.55\,\rho_b\,\omega_{\text{SOM}}$ [$\text{m}^3\,\text{m}^{-3}$] [Bogena et al., 2020]; and (ii) above-ground biomass water equivalent AGBH $\approx$ LAI $\times$ 150 $\text{g}\,\text{m}^{-2}$ [Baatz et al., 2015, Schrön et al., 2017].

2.3 Topographic Correction κ_{topo}

A CRNS probe on flat, homogeneous terrain sees a reference soil volume $V_0 = \pi r_{86}^2 z_{86}$. Real terrain deviates from this ideal, either by presenting extra soil (terrain above sensor level) or reduced soil (valleys, cliffs). The topographic correction is defined as

$$\kappa_{\text{topo}} = \frac{V_{\text{eff}}}{V_0}, \tag{7}$$

where V_{eff} is the $W(r)$-weighted effective soil volume actually present.

The pipeline implements a *3-D ray-casting* algorithm. For each ray emitted from the sensor at zenith angle θ and azimuth ϕ, the algorithm computes:

1. The air path length L_{air} from sensor to the first DEM intersection, giving the attenuation factor $\exp(-L_{\text{air}}/\lambda_{\text{air}})$ with $\lambda_{\text{air}} = 130\,\text{g}\,\text{cm}^{-2} \,/\, \rho_{\text{air}}$.
2. The soil path length $L_{\text{soil}} = z_{86}/\cos\alpha$, corrected for the local surface slope angle α via the DEM gradient, giving the production term $1 - \exp(-L_{\text{soil}}/\lambda_{\text{soil}})$ with $\lambda_{\text{soil}} = 162\,\text{g}\,\text{cm}^{-2} \,/\, \rho_b$.
3. The angular weight $\cos^2\theta$, accounting for both the Lambert emission pattern and the solid-angle element $\mathrm{d}\Omega = \cos\theta\,\mathrm{d}\theta\,\mathrm{d}\phi$.

The ray contribution is

$$\mathrm{d}N \propto \exp\!\left(-\tfrac{L_{\text{air}}}{\lambda_{\text{air}}}\right)\left[1 - \exp\!\left(-\tfrac{L_{\text{soil}}}{\lambda_{\text{soil}}}\right)\right]\cos^2\theta\,, \tag{8}$$

and κ_{topo} is the ratio of the sum over all DEM rays to the sum over an equivalent flat reference surface. $\kappa_{\text{topo}} > 1$ indicates excess terrain (over-estimation of SM without correction); $\kappa_{\text{topo}} < 1$ indicates missing terrain (under-estimation).

2.4 Muon Field-of-View Correction κ_{muon}

High-energy cosmic muons arriving at the surface contribute to the secondary neutron production via spallation reactions in soil. Their differential flux follows $\mathrm{d}N/\mathrm{d}\Omega \propto \cos^n\theta_z$ with $n \approx 2$. Surrounding terrain that rises above the horizon at elevation angle $\psi(\phi)$ blocks all muons with zenith angle $\theta_z > 90^\circ - \psi(\phi)$, reducing the effective muon flux.

The muon correction is defined as the ratio of the visible-sky integral to the full-hemisphere integral:

$$\kappa_{\text{muon}} = \frac{\displaystyle\int_0^{2\pi}\int_0^{\theta_{z,\max}(\phi)} \cos^2\theta_z \sin\theta_z\,\mathrm{d}\theta_z\,\mathrm{d}\phi}{\displaystyle\int_0^{2\pi}\int_0^{\pi/2} \cos^2\theta_z \sin\theta_z\,\mathrm{d}\theta_z\,\mathrm{d}\phi}, \tag{9}$$

where $\theta_{z,\max}(\phi) = \pi/2 - \psi(\phi)$. The azimuthal integral evaluates analytically per bin:

$$f(\phi) = \frac{1 - \cos^3 \theta_{z,\max}(\phi)}{3}. \tag{10}$$

By construction $\kappa_{\text{muon}} \leq 1$; it equals 1 on a flat site with no surrounding topography.

2.5 Land-Use/Land-Cover Correction κ_{lulc}

Non-soil hydrogen sources within the footprint — vegetation water, standing water, built structures — modulate neutron moderation. Following Schrön et al. [2017] and Iwema et al. [2017], each land-cover class c is assigned a hydrogen equivalence factor $f_H^{(c)}$, normalised to bare soil at the reference moisture θ_v^{ref}. The correction is

$$\kappa_{\text{lulc}} = \frac{\sum_{i \in \text{footprint}} W(r_i)\, f_H^{(c_i)}\, A_i}{\sum_{i \in \text{footprint}} W(r_i)\, A_i}, \tag{11}$$

where A_i is the pixel area and c_i its WorldCover class. Selected f_H values are listed in Table 1.

Table 1: Hydrogen equivalence factors f_H by land cover class (relative to bare soil baseline $f_H = 1$). Sources: Schrön et al. [2017], Iwema et al. [2017], Franz et al. [2012].

Land-cover class	f_H	Notes
Water / wetland	4.0 / 2.5	$\approx 4\times$ soil H
Snow / glacier	3.0	
Tree cover (forest)	1.5	includes canopy water
Shrubland	1.15	
Grassland / cropland	0.95–1.00	
Built-up	0.05	concrete, asphalt
Bare / sparse veg.	0.03	

For the OSM vector layer the baseline contribution of uncovered footprint area is computed as $f_H^{\text{base}} \times (\Sigma_{\text{DEM}} - \Sigma_{\text{OSM}})$ where both sums are $W(r)$-weighted integrals, avoiding the spatially averaged approximation used in simpler implementations.

2.6 Site Flux Estimates and N_0

The expected epithermal neutron count rate at the site is

$$N_{\text{neut,site}} = N_{\text{sl}}\, f_{R_c}\, \exp[\beta_n\,(P_0 - P)]\, \kappa_{\text{topo}}, \tag{12}$$

where N_{sl} is the detector-specific sea-level reference count, $f_{R_c} = \exp[\alpha_{\text{H}}(R_c^{\text{site}} - R_c^{\text{ref}})]$ is the geomagnetic rigidity correction [Hawdon et al., 2014] with $\alpha_{\text{H}} = -0.075\,\text{GV}^{-1}$, and $\beta_n = 0.0077\,\text{hPa}^{-1}$ is the barometric coefficient for neutrons. The site pressure P follows the standard exponential atmosphere with scale height $H = 8500\,\text{m}$. Rigidity cutoff R_c is retrieved from the Smart & Shea (2019) tables via `crnpy`.

The full correction chain is

$$\kappa_{\text{tot}} = \kappa_{\text{topo}} \cdot \kappa_{\text{muon}} \cdot \kappa_{\text{lulc}}, \tag{13}$$

and N_0 follows from Eq. (6).

3 Data Sources

All datasets used by the pipeline are freely accessible from global open repositories. Table 2 lists each source with the relevant variables, spatial resolution, and access method. No registration is required for any source; all downloads are executed automatically at run time through standard HTTP requests.

Table 2: Data sources used by the CRNS site characterization pipeline.

Source	Variables	Resolution	Access
Copernicus DEM GLO-30	Elevation (m a.s.l.)	30 m	AWS S3 (open)
ESA WorldCover 2021 v200	Land cover class (11 classes)	10 m	AWS S3 / Planetary Compu
OpenStreetMap (Overpass API)	Water bodies, roads, buildings	Vector	REST API
SoilGrids 2.0	Sand, silt, clay (%), bulk density, SOC	250 m	ISRIC REST API
ERA5-Land / Open-Meteo Archive	Temperature, RH, snowfall, snow depth (daily, 5 yr); soil moisture 0–100 cm (hourly, $\geq$2015)	~9 km (soil moisture) / ~30 km (climate)	Open-Meteo REST API
MODIS MOD15A2H	Leaf Area Index (LAI), 8-day composites	500 m	NASA AppEEARS / pystac
Macrostrat API v2	Geologic units, lithology, age	Global (variable)	Macrostrat REST API
Smart and Shea [2009] / `crnpy`	Geomagnetic rigidity cutoff R_c	1° × 1° grid	Python package

3.1 Digital Elevation Model

The Copernicus DEM GLO-30 [European Space Agency and Airbus, 2022] provides a near-global 30-m resolution surface elevation model derived from TanDEM-X radar interferometry. The pipeline downloads the tiles covering a square bounding box $[-r_{\mathrm{buf}}, +r_{\mathrm{buf}}]$ centred on the site, with default buffer $r_{\mathrm{buf}} = 3\,r_{86} \approx 350\,\mathrm{m}$. Tiles are mosaicked and reprojected to a metric coordinate system (UTM, WGS-84) before use.

The DEM is the master grid: all raster products are resampled to match its spatial extent and pixel spacing.

3.2 Land-Use / Land-Cover

ESA WorldCover 2021 [Zanaga et al., 2022] provides a 10-m global land-cover classification based on Sentinel-1 and Sentinel-2 imagery. The 11 classes (tree cover, shrubland, grassland, cropland, built-up, bare/sparse vegetation, snow/ice, permanent water, herbaceous wetland, mangroves, moss/lichen) are aggregated to the hydrogen equivalence groups of Table 1.

OpenStreetMap (OSM) features — surface water bodies, roads with sealed surfaces, and building footprints — are queried via the Overpass API within the $2\,r_{86}$ radius of the footprint. OSM vectors complement WorldCover for fine-scale features (individual buildings, canals) that may be below the 10-m raster resolution.

The JRC Global Surface Water dataset [Pekel et al., 2016] provides occurrence frequency (0–100%) for permanent and seasonal surface water, used to distinguish permanent from temporary water bodies in the f_H assignment.

3.3 Soil Properties

SoilGrids 2.0 [Poggio et al., 2021] provides global predictions at six depth intervals (0–5, 5–15, 15–30, 30–60, 60–100, 100–200 cm) for:

- sand, silt, clay fractions $[\mathrm{g\,kg^{-1}}]$;
- bulk density (fine earth) $[\mathrm{cg\,cm^{-3}}]$;
- soil organic carbon (SOC) content $[\mathrm{dg\,kg^{-1}}]$.

The pipeline uses the 0–30 cm depth-weighted mean, which matches the typical z_{86} penetration range. Lattice water l_w is derived from the clay fraction using the empirical relation of Köhli et al. [2021]:

$$l_w = 0.097\,f_{\mathrm{clay}} + 0.033, \tag{14}$$

where f_{clay} is the clay mass fraction [g/g].

3.4 Atmospheric Data

Monthly climatologies of near-surface air temperature, relative humidity, snowfall, and snow depth are downloaded from the Open-Meteo ERA5 archive [Hersbach et al., 2020] for the 5 years preceding the analysis date. Daily values are aggregated to monthly means and then averaged across years to obtain a 12-month climatology.

From temperature T [°C] and relative humidity RH [%], the atmospheric water-vapour density ρ_{WV} $[\mathrm{g\,m^{-3}}]$ is computed via the Magnus formula:

$$e_{\mathrm{sat}}(T) = 611.2\exp\left(\frac{17.67\,T}{T + 243.5}\right) \quad [\mathrm{Pa}], \tag{15}$$

$$\rho_{WV} = \frac{\mathrm{RH}}{100}\,\frac{e_{\mathrm{sat}}}{R_v\,T_K} \quad [\mathrm{g\,m^{-3}}], \tag{16}$$

where $R_v = 461.5\,\mathrm{J\,kg^{-1}\,K^{-1}}$ is the specific gas constant for water vapour and T_K is temperature in Kelvin.

Snow-water equivalent (SWE) is estimated as

$$\mathrm{SWE} = h_{\mathrm{snow}} \times \rho_{\mathrm{snow}}, \tag{17}$$

with h_{snow} the mean snow depth [m] and $\rho_{\mathrm{snow}} = 250\,\mathrm{kg\,m^{-3}}$ a representative seasonal density for wet alpine snow.

3.5 Soil Moisture — ERA5-Land

Hourly volumetric soil moisture is retrieved from the ERA5-Land reanalysis via the Open-Meteo Historical API. The module `era5sm.py` downloads three depth layers: 0–7 cm, 7–28 cm, and 28–100 cm, all in $\mathrm{m^3\,m^{-3}}$. Downloads proceed year by year from a configurable `start_year` (default 2015) to the current date, with each annual time-series stored as a compressed `.npz` file (`hourly_YYYY.npz`). Monthly statistics (mean, standard deviation, minimum, maximum, and observation count) are derived from the full multi-year record and cached in a single `monthly_agg.npz`. On subsequent runs only missing years are downloaded; the monthly aggregates are recomputed only if the set of cached years has changed.

The three ERA5-Land depth layers bracket the typical CRNS penetration depth z_{86}, making the 0–7 cm layer the primary reference for the data-fusion step (Section 4.7).

3.6 Vegetation (LAI)

The MODIS MOD15A2H product provides 8-day, 500-m composite estimates of leaf area index (LAI) [$\mathrm{m^2\,m^{-2}}$]. Monthly means are derived from all cloud-free composites within the analysis year. LAI is used to estimate above-ground biomass hydrogen (AGBH) as described in Section 4.5.

3.7 Geology

The Macrostrat API v2 provides global geological map data at multiple resolution scales ("large", "medium", "small"). For each query point the pipeline retrieves the dominant lithological unit(s), their Macrostrat rock type, geologic age, and estimated hydrogen inventory impact on l_w. Where lithological data are unavailable (ocean, ice sheet), the pipeline returns a "not available" flag and uses the SoilGrids-derived l_w without geological supplement.

4 Computational Methods

4.1 Pipeline Overview

The pipeline executes sequentially in 16 numbered steps. Each step caches its result to disk (JSON or NumPy `.npz`); on re-run with identical inputs only missing steps are recomputed. The high-level sequence is:

1. Download and mosaic Copernicus DEM tiles.
2. Compute terrain derivatives (slope, aspect, horizon profile, TWI — cached).
3. Download ESA WorldCover and OSM vector features.
4. Compute *preliminary* r_{86} and z_{86} at default $\theta_v = 0.20$ and $l_w = 0$.
5. Download SoilGrids soil properties for the footprint.
6. Compute actual l_w from clay fraction (Eq. (14)), recompute z_{86} and r_{86}.
7. Compute κ_{topo} (3-D ray-casting over DEM).
8. Compute κ_{muon} (horizon-based muon integral).
9. Compute κ_{lulc} (WorldCover + OSM, $W(r)$-weighted).
10. Compute site neutron flux and N_0 (Eqs. (12)–(6)).
11. Download ERA5 climatology and compute ρ_{WV}, f_{WV}, SWE (monthly).
12. Download MODIS LAI and compute AGBH.
13. Query Macrostrat for geological characterisation.
14. Download ERA5-Land hourly soil moisture (year by year, cached); compute multi-year monthly statistics.
15. Downscale and fuse ERA5 soil moisture with local pedology, topography, and land cover; propagate uncertainty.
16. Compute optimal soil sampling plan and r_{86} dynamic range.
17. Assemble the full report (PDF + JSON).

4.2 Terrain Processing

Slope and aspect. Slope β and aspect are derived from the DEM by the standard central-difference gradient:

$$\nabla z = \left(\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}\right), \qquad \beta = \arctan|\nabla z|. \tag{18}$$

A minimum slope floor of 0.01° is applied to avoid division by zero in subsequent hydrological calculations.

Horizon profile. For each of $N_\phi = 72$ azimuth directions (5° spacing), the maximum elevation angle $\psi(\phi)$ is computed by scanning along the DEM at that azimuth out to $3\,r_{86}$ from the site. The result is a 72-element array $\psi(\phi_k)$ that enters both the κ_{muon} integral and the ray-casting algorithm.

Topographic Wetness Index. TWI is computed as [Beven and Kirkby, 1979]:

$$\text{TWI} = \ln \frac{A_s}{\tan\beta}, \tag{19}$$

where A_s is the specific upslope drainage area per unit contour length, derived from the D_∞ flow accumulation of the pit-filled DEM [Wang and Liu, 2006]. Because TWI computation is the most expensive terrain step ($\mathcal{O}(N_{\text{pix}} \log N_{\text{pix}})$ for large DEMs), results are cached to disk as a `.npz` file. The cache key is a hash of the DEM shape, pixel spacing, r_{86}, and summary statistics (mean and standard deviation of elevation); if the hash matches the stored file the cached result is returned directly.

4.3 Topographic Correction κ_{topo}

The 3-D ray-casting implementation samples $N_\theta = 18$ zenith angles (uniformly in $\cos\theta$, 0°–89°) and $N_\phi = 72$ azimuth angles. For each ray (θ_k, ϕ_j):

1. March along the ray in 3-D until it intersects the DEM surface, recording the air path length L_{air}.
2. At the intersection, read the surface slope α and compute $L_{\text{soil}} = z_{86}/\cos\alpha$.
3. Evaluate Eq. (8).

The flat-reference integral uses a perfectly horizontal DEM at the sensor elevation. Both integrals are computed with the same angular sampling; κ_{topo} is their ratio.

Rays that exit the DEM bounding box without hitting terrain (i.e., the site is on a ridge or cliff edge) contribute zero to the numerator but retain their full flat-reference contribution in the denominator, correctly reducing κ_{topo} below 1.

4.4 LULC Correction and Supplementary Hydrogen

WorldCover raster. Each 10-m WorldCover pixel within $2\,r_{86}$ of the site is assigned the hydrogen equivalence factor $f_H^{(c)}$ from Table 1. The radial weight $W(r_i)$ (Eq. (3)) is evaluated at the pixel centroid distance r_i. The correction factor κ_{lulc} follows Eq. (11).

OSM vector features. OSM polygons (buildings, water bodies, sealed roads) are rasterised onto the DEM grid at 1-m resolution and then downsampled to the DEM pixel size. For pixels with OSM overlap the WorldCover f_H is replaced by the OSM-specific value (e.g. $f_H = 4.0$ for permanent water, $f_H = 0.05$ for sealed surfaces). The uncovered area inside the $2\,r_{86}$ circle contributes

$$\text{base_contrib} = f_H^{\text{base}} \times \max\!\left(0,\ \textstyle\sum_i W(r_i)\,A_i - \sum_{j\in\text{OSM}} W(r_j)\,A_j\right), \tag{20}$$

where f_H^{base} is the WorldCover bare-soil baseline ($f_H = 0.03$) and the difference of W-weighted integrals avoids the spatially averaged approximation used in simpler implementations.

Reference moisture. All f_H values and the WorldCover baseline are computed at the user-specified initial soil moisture θ_v^{ref} (`THETA_V_INIT` in the configuration file, default $0.20\,\text{m}^3\,\text{m}^{-3}$). This makes κ_{lulc} physically consistent with the r_{86} and z_{86} computed at the same moisture level.

4.5 Supplementary Correction Factors

These factors are not included in κ_{tot} but are reported separately as operational correction guidance.

Atmospheric water vapour. An increase in above-sensor water vapour reduces the epithermal neutron flux reaching the ground. Following Rosolem et al. [2013], the correction factor is:

$$f_{WV} = \exp\!\left(-\alpha_{WV}\,\rho_{WV}\right), \tag{21}$$

with $\alpha_{WV} = 0.0054\,\text{m}^3\,\text{g}^{-1}$ [Zreda et al., 2012]. Monthly climatological values of ρ_{WV} (Eqs. (15)–(16)) and the corresponding f_{WV} are tabulated in the report. Note that f_{WV} is an operational (epoch-by-epoch) correction; the climatology here serves as site-characterisation guidance.

Above-ground biomass hydrogen (AGBH). Vegetation intercepts and stores water; the associated hydrogen moderates neutrons [Baatz et al., 2015, Schrön et al., 2017]. An effective soil-moisture equivalent is estimated as:

$$\mathrm{AGBH} = \mathrm{LAI} \times 150\,\mathrm{g\,m^{-2}} + \theta_{\mathrm{litter}} \quad [\mathrm{mm}], \tag{22}$$

where the 150 g m^{-2} factor is the mean interception capacity per unit LAI [Schrön et al., 2017] and θ_{litter} is an optional litter-water depth.

Snow-water equivalent. Snow on the ground effectively acts as liquid water from the neutron probe's perspective. Monthly mean SWE (Eq. (17)) is reported alongside the month in which snow cover first exceeds 150 mm SWE — a threshold at which the detector is largely opaque to cosmic-ray neutrons and calibration is impractical [Bogena et al., 2020].

Soil organic carbon hydrogen equivalent. Soil organic matter (SOM) contains hydrogen at a concentration similar to liquid water. The volumetric hydrogen equivalent is approximated as [Bogena et al., 2020]:

$$\theta_{\mathrm{SOC}} \approx 0.55\,\rho_b\,\omega_{\mathrm{SOM}} \quad [\mathrm{m^3\,m^{-3}}], \tag{23}$$

where ω_{SOM} is the SOM mass fraction [g/g] (SOM $\approx 1.724\times$ SOC by convention). This term is reported alongside l_w as a supplementary offset that does not change with time.

4.6 Optimal Soil Sampling Plan

A CRNS calibration requires in-situ gravimetric soil moisture measurements distributed across the footprint in proportion to the neutron sensitivity kernel $W(r)$. The pipeline designs a plan of $n_{\mathrm{tot}} = 26$ sampling locations arranged in four concentric rings at radii $r_k = \{0.25, 0.50, 0.75, 1.00\} \times r_{86}$.

Ring weight. The contribution of ring k to the footprint integral is

$$\Phi_k = \int_{r_{k-1}}^{r_k} W(r)\,2\pi r\,\mathrm{d}r, \tag{24}$$

evaluated numerically by the trapezoidal rule. Points per ring $n_k \in \{4, 6, 8, 8\}$ are allocated so that $n_k / \sum n_j \approx \Phi_k / \sum \Phi_j$.

Sample depth. At each location, soil is sampled at three depth intervals (0–5, 5–15, 15–30 cm) to resolve the vertical weighting of neutron penetration.

Footprint dynamic range. As supplementary information, the report includes a plot of $r_{86}(\theta_v)$ computed over the full moisture range [0.01, 0.50] m^3 m^{-3}. This shows how the footprint radius, and hence the spatial representativeness of the measurements, changes between dry and wet conditions at the site.

4.7 Soil Moisture Downscaling and Data Fusion

ERA5-Land soil moisture has a native resolution of ∼9 km and is produced with a globally uniform land-surface model. Local conditions (topography, soil texture, land cover) can shift the true soil moisture at the CRNS site significantly relative to the ERA5 grid-cell value. The module `smphysics.py` applies a sequential chain of physically motivated corrections implemented in `fuse_soil_moisture()`.

Step 1 — ERA5 raw signal. The 0–7 cm monthly mean from `era5sm.py` is used as the starting estimate, as this layer most closely matches the effective sensing depth of CRNS at low to moderate moisture.

Step 2 — Pedological rescaling (Saxton & Rawls 2006). ERA5-Land uses globally uniform hydraulic parameters. Site-specific Field Capacity (FC) and Wilting Point (WP) are derived from the SoilGrids sand, clay, and organic-matter fractions via the pedotransfer functions of Saxton and Rawls [2006]:

$$\theta_{\text{local}} = \text{WP}_{\text{loc}} + \frac{\theta_{\text{ERA5}} - \text{WP}_{\text{ERA5}}}{\text{FC}_{\text{ERA5}} - \text{WP}_{\text{ERA5}}} \left(\text{FC}_{\text{loc}} - \text{WP}_{\text{loc}}\right), \tag{25}$$

with $\text{FC}_{\text{ERA5}} = 0.32$ and $\text{WP}_{\text{ERA5}} = 0.12\,\text{m}^3\,\text{m}^{-3}$ as representative global defaults. The result is clipped to $[\text{WP}_{\text{loc}}, \text{FC}_{\text{loc}}]$.

Step 3 — Topographic correction. Two terrain effects are applied:

1. *TWI term (additive):* sites with TWI above the footprint mean accumulate more lateral flow and are therefore wetter:

$$\Delta\theta_{\text{TWI}} = \beta_{\text{TWI}} \left(\text{TWI}_{\text{site}} - \overline{\text{TWI}}_{\text{fp}}\right), \tag{26}$$

   with default $\beta_{\text{TWI}} = 0.008\,\text{m}^3\,\text{m}^{-3}$ per TWI unit, clipped to ± 0.15.
2. *PISR term (multiplicative):* higher annual solar irradiation increases evapotranspiration and dries the soil:

$$f_{\text{PISR}} = \exp\left(-\alpha_{\text{PISR}} \frac{\text{PISR}_{\text{site}} - \text{PISR}_{\text{ERA5}}}{\text{PISR}_{\text{ERA5}}}\right), \tag{27}$$

   with default $\alpha_{\text{PISR}} = 0.12$.

The combined topographic correction is $\theta_{\text{topo}} = (\theta_{\text{local}} + \Delta\theta_{\text{TWI}}) \times f_{\text{PISR}}$.

Step 4 — LULC correction. A categorical correction is applied based on WorldCover class fractions and the κ_{LULC} already computed for the neutron sensitivity model:

- Snow/ice $> 30\%$: $\theta \times 0.4$ (snow is not liquid water);
- Impervious (built-up + bare) $> 50\%$: $\theta \times 0.30$;
- Impervious 20–50%: $\theta \times (1 - 0.5\, f_{\text{imp}})$;
- Wetland + forest $> 40\%$ and $\kappa_{\text{LULC}} > 1.2$: $\theta \times 1.10$ (structurally wet surface).

Uncertainty. A total uncertainty is propagated in quadrature from four independent components (Gaussian error propagation):

$$\sigma_{\text{total}} = \sqrt{\sigma^2_{\text{ERA5}} + \sigma^2_{\text{topo}} + \sigma^2_{\text{pedol}} + \sigma^2_{\text{LULC}}} \approx 0.066\ \text{m}^3\,\text{m}^{-3}, \tag{28}$$

with default components $\sigma_{\text{ERA5}} = 0.05$, $\sigma_{\text{topo}} = 0.02$, $\sigma_{\text{pedol}} = 0.03$, $\sigma_{\text{LULC}} = 0.02$. The report provides monthly $\theta_v \pm \sigma$ bounds alongside the ERA5 raw signal.

4.8 Geological Characterisation

The Macrostrat API v2 is queried with the site coordinates at three resolution scales (large, medium, small) in descending order of coverage. The first non-empty response is used. The pipeline extracts:

- dominant lithology name and Macrostrat rock type (carbonate, siliciclastic, igneous, ...);
- estimated influence on l_w (low, medium, high), based on the known hydrogen content of the rock type;
- an indicative radiogenicity level (for U/Th-based radiation background assessment);
- geologic age era (Cenozoic, Mesozoic, Palaeozoic, Precambrian).

This qualitative summary complements the quantitative SoilGrids l_w estimate when geological context is relevant (e.g. evaporite-rich or serpentinised terrain).

5 Worked Example: Italian Alps Test Site

To illustrate the pipeline output, we apply it to a candidate installation site in the Italian Alps: a sub-alpine meadow at approximately 46.4° N, 11.3° E, elevation 1850 m a s l The site is representative of the challenging conditions that motivate careful pre-installation characterisation: significant relief, mixed land cover (meadow, sparse conifers, rocky outcrops), seasonal snow, and clay-rich soils derived from carbonate–flysch geology.

5.1 Terrain and Topographic Correction

Figure 1 shows the 30-m Copernicus DEM mosaicked over the ±500 m bounding box around the sensor location. The terrain is characterised by a north-facing slope ($\sim 8°$ mean slope within r_{86}) with a prominent ridge to the south-east rising approximately 120 m above the sensor.

Figure 1: Hillshaded DEM with the r_{86} (inner circle) and $3\,r_{86}$ (outer circle) footprint boundaries overlaid. The sensor location is marked with a cross.

The 3-D ray-casting algorithm over 18×72 directions gives

$$\kappa_{\text{topo}} = 0.97,$$

indicating that the south-eastern ridge reduces the effective soil volume by approximately 3% relative to a flat reference. Without this correction, the Desilets N_0 would be overestimated by the same margin.

5.2 Muon Field-of-View Correction

The horizon profile $\psi(\phi)$ shows a maximum elevation angle of 8.5° towards azimuth 145° (south-south-east) and is below 3° for all other azimuths. The analytical κ_{muon} integral (Eq. (9)) gives

$$\kappa_{\text{muon}} = 0.992,$$

a 0.8% reduction in the effective muon flux. This is small but non-negligible at a site with high-precision calibration requirements.

5.3 Land-Use / Land-Cover Correction

Figure 2 shows the ESA WorldCover 2021 map reprojected to the DEM grid. Within the r_{86} circle the land cover is dominated by grassland ($f_H = 0.95$, 64 % of weighted area), tree cover ($f_H = 1.50$, 18 %), and bare rock ($f_H = 0.03$, 11 %). A small seasonal stream mapped in OSM ($f_H = 4.0$) covers 0.4 % of the $W(r)$-weighted area.

The weighted κ_{lulc} (Eq. (11)) evaluates to

$$\kappa_{\text{lulc}} = 1.08,$$

Figure 2: ESA WorldCover 2021 classification within the $2\,r_{86}$ footprint circle, with OSM water features overlaid. The r_{86} boundary is the solid white circle.

reflecting the net positive hydrogen contribution from tree canopy water relative to the bare-soil baseline. Without this correction, the calibrated N_0 would be overestimated by $\sim 8\%$, propagating directly into a soil-moisture bias of $\sim 0.02\,\mathrm{m}^3\,\mathrm{m}^{-3}$.

5.4 Total Correction and Reference Count

The total correction factor is

$$\kappa_{\mathrm{tot}} = \kappa_{\mathrm{topo}} \cdot \kappa_{\mathrm{muon}} \cdot \kappa_{\mathrm{lulc}} = 0.97 \times 0.992 \times 1.08 = 1.039.$$

With a detector-specific sea-level reference count $N_{\mathrm{sl}} = 2800\,\mathrm{cph}$ and a site rigidity cutoff $R_c = 4.2\,\mathrm{GV}$ (retrieved from `crnpy`):

$$f_{R_c} = \exp[\alpha_H(R_c^{\mathrm{site}} - R_c^{\mathrm{ref}})] = \exp[-0.075 \times (4.2 - 4.49)] = 1.022,$$
$$N_{\mathrm{neut,site}} = 2800 \times 1.022 \times \exp[0.0077 \times (1013.25 - 794)] \times 1.039 = 3620\,\mathrm{cph}.$$

With $l_w = 0.055$ g/g (from SoilGrids clay fraction $f_{\mathrm{clay}} = 0.23$, Eq. (14)):

$$N_0 = \frac{3620}{0.0808 + 0.372\,/\,(0.055 + 0.115)} = \frac{3620}{0.0808 + 2.188} \approx 1601\,\mathrm{cph}.$$

This value enters the Desilets inversion (Eq. (5)) for all subsequent soil-moisture retrievals.

5.5 Soil Properties and Supplementary Hydrogen Inventory

Table 3 summarises the SoilGrids-derived soil properties for the 0–30 cm layer.

Table 3: Soil properties at the Italian Alps test site (0–30 cm depth-weighted mean, SoilGrids 2.0).

Property	Value	Unit
Bulk density ρ_b	1.18	$\mathrm{g\,cm^{-3}}$
Clay fraction f_{clay}	0.23	g/g
SOC content	28	$\mathrm{g\,kg^{-1}}$
Lattice water l_w	0.055	g/g
θ_{SOC}	0.028	$\mathrm{m^3\,m^{-3}}$
z_{86} (at $\theta_v = 0.20$)	26	cm

5.6 Atmospheric and Vegetation Seasonal Climatology

Figure 3 shows the monthly climatological ρ_{WV}, f_{WV}, SWE and LAI averaged over 2019–2023.

Key observations:

Figure 3: Monthly climatology at the Italian Alps test site. *Top left*: atmospheric water-vapour density ρ_{WV} [$\mathrm{g\,m^{-3}}$]. *Top right*: neutron correction factor f_{WV}. *Bottom left*: snow-water equivalent SWE [mm]. *Bottom right*: MODIS LAI.

- ρ_{WV} ranges from $\sim 3\,\mathrm{g\,m^{-3}}$ in January to $\sim 12\,\mathrm{g\,m^{-3}}$ in July, implying an operational f_{WV} variation of 5–6% between winter and summer.
- SWE exceeds the 150 mm opacity threshold in January–March, making those months unsuitable for soil-moisture calibration.
- Peak LAI in July ($\approx 3.2\ \mathrm{m^2/m^2}$) gives AGBH $\approx 480\,\mathrm{mm}$, equivalent to $\sim 0.048\,\mathrm{m^3\,m^{-3}}$ additional hydrogen — an important bias term in summer.

5.7 Optimal Sampling Plan

Figure 4 shows the recommended 26-point sampling plan superimposed on the DEM and the $r_{86}(\theta_v)$ dynamic-range curve.

Figure 4: *Left*: recommended soil sampling locations (filled circles, coloured by ring) overlaid on the hillshaded DEM. Ring radii are $\{0.25, 0.50, 0.75, 1.00\} \times r_{86}$. *Right*: footprint radius r_{86} as a function of volumetric soil moisture θ_v at site pressure. Dashed line marks the design moisture $\theta_v = 0.20$.

At the design moisture of $\theta_v = 0.20\,\mathrm{m^3\,m^{-3}}$, $r_{86} = 113\,\mathrm{m}$. The four sampling rings are therefore located at $r = 28\,,\,57\,,\,85\,,\,113\,\mathrm{m}$. The inner ring (4 points) carries $\Phi_1/\sum\Phi = 18\%$ of the total footprint weight; the outer ring (8 points) carries 32%. The dry-soil extreme ($\theta_v \to 0$) gives $r_{86} \to 188\,\mathrm{m}$; at saturation ($\theta_v = 0.45$) r_{86} shrinks to 106 m, a 44% dynamic range.

6 Limitations and Uncertainties

6.1 DEM Resolution

The 30-m Copernicus DEM GLO-30 is adequate for computing κ_{topo} and κ_{muon} at most CRNS sites, where the footprint radius exceeds 100 m. However, features smaller than ∼60 m (two pixels) are not resolved: narrow gullies, embankments, road cuts, and individual buildings may be missed. In complex built or agricultural terrain, a higher-resolution LiDAR DEM (1–5 m) is recommended. The ray-casting algorithm accepts any DEM; swapping the 30-m tile for a LiDAR product requires only a change in the download module.

The horizon-scan algorithm is limited to azimuths sampled at 5° intervals (72 directions). Narrow topographic gaps (e.g. a mountain pass at 4° width) between two blocking ridges may be missed. Increasing N_ϕ reduces this error at the cost of proportionally more CPU time.

6.2 LULC Classification

The ESA WorldCover 2021 product has a reported overall accuracy of ∼75% at the 10-m pixel level. Classification errors in the heterogeneous environments common to CRNS sites (forest edges, rice paddies, urban green space) can propagate directly into κ_{lulc}. A 10% misclassification of forest/grassland pixels corresponds to a $\Delta\kappa_{\text{lulc}} \approx 0.06$ error, translating to a soil-moisture bias of approximately $0.006\,\text{m}^3\,\text{m}^{-3}$.

The f_H values in Table 1 are literature means; actual vegetation water content varies with phenology and rainfall. For precise calibration, site-specific leaf water content measurements at the time of calibration are preferable.

6.3 Soil Properties

SoilGrids 2.0 provides 90th-percentile uncertainty maps alongside its predictions. Bulk density ρ_b and clay fraction have typical uncertainties of ± 0.1 g/cm^3 and ± 5 g/kg respectively at the 250-m resolution. These propagate into l_w and z_{86}:

$$\delta l_w \approx 0.097\,\delta f_{\text{clay}} \sim 0.005 \text{ g/g},$$
$$\delta z_{86} \approx \frac{8.3}{\rho_b^2(0.0564 + \theta_v + l_w)}\,\delta\rho_b \sim 2\,\text{cm}.$$

The SoilGrids uncertainty is a spatial mean at 250 m; real local variability, especially in organic-rich or heterogeneous soils, can be larger. In-situ soil sampling at the calibration rings (Sec. 4.6) remains the gold standard for ρ_b and clay determination.

6.4 Lattice Water Estimate

Equation (14) is an empirical fit across a global soil database. For soils dominated by 1:1 clays (kaolinite, halloysite) the coefficient 0.097 may overestimate l_w by 20–30%, whereas for 2:1 swelling clays (smectite, montmorillonite) the same equation may underestimate it. Geological characterisation (Sec. 4.6) provides a qualitative flag when the site lithology suggests an unusual clay mineralogy.

6.5 Atmospheric Corrections

The ERA5-derived ρ_{WV} climatology has a spatial resolution of ∼30 km and may not capture local valley inversions, sea-breeze effects, or orographic enhancement of humidity. For sites with strong mesoscale meteorological forcing, on-site humidity measurements are recommended for operational f_{WV} correction.

The SWE estimate uses a fixed snow density $\rho_{\text{snow}} = 250\,\text{kg}\,\text{m}^{-3}$. Fresh snow and densified spring snowpack can deviate by $\pm 100\,\text{kg}\,\text{m}^{-3}$, introducing a 40% uncertainty in SWE. This affects only the seasonal operability assessment (month of SWE $> 150\,\text{mm}$) and not the soil-moisture retrievals.

6.6 Geological Data

The Macrostrat API provides a qualitative, reconnaissance-level geological characterisation. At some sites — particularly in data-sparse regions (parts of Africa, Central Asia, Antarctica) — no unit may be returned, and the fallback is a "not available" flag. Hydrogeological properties (fracturing, karst porosity, perched water tables) cannot be retrieved and must be assessed through field investigation.

6.7 Geomagnetic Rigidity

The rigidity cutoff R_c from the Smart & Shea (2019) tables carries an uncertainty of ~ 0.1 GV [Smart and Shea, 2009], which translates to $\Delta f_{R_c} \approx 0.7\%$. This is typically the smallest uncertainty in the N_0 chain and can be neglected in most applications.

6.8 Combined Uncertainty on N_0

Assuming the dominant sources are independent and normally distributed, the relative uncertainty on N_0 is approximately

$$\frac{\sigma_{N_0}}{N_0} \approx \sqrt{\left(\frac{\sigma_{\kappa_{\text{topo}}}}{\kappa_{\text{topo}}}\right)^2 + \left(\frac{\sigma_{\kappa_{\text{muon}}}}{\kappa_{\text{muon}}}\right)^2 + \left(\frac{\sigma_{\kappa_{\text{lulc}}}}{\kappa_{\text{lulc}}}\right)^2 + \left(\frac{\sigma_{l_w}}{l_w + a_2}\right)^2}. \tag{29}$$

For the Italian Alps example, a rough budget gives: $\sigma_{\kappa_{\text{topo}}}/\kappa_{\text{topo}} \approx 1\%$ (DEM resolution), $\sigma_{\kappa_{\text{muon}}}/\kappa_{\text{muon}} \approx 0.2\%$ (horizon sampling), $\sigma_{\kappa_{\text{lulc}}}/\kappa_{\text{lulc}} \approx 4\%$ (LULC classification + f_H), $\sigma_{l_w}/(l_w + a_2) \approx 3\%$ (SoilGrids clay uncertainty). This gives $\sigma_{N_0}/N_0 \approx 5\%$, corresponding to a soil-moisture uncertainty of roughly $0.015\,\text{m}^3\,\text{m}^{-3}$ at $\theta_v = 0.20\,\text{m}^3\,\text{m}^{-3}$. This is comparable to the precision of a single calibration measurement and underscores the value of the site characterisation workflow.

7 Conclusions

We have presented the *CRNS Site Characterization Pipeline*, a fully integrated Python tool for automating the pre-installation assessment of Cosmic-Ray Neutron Sensing stations from global open datasets. The pipeline addresses a recognised gap in the CRNS ecosystem: while operational correction routines are available (e.g. `crnpy`), no single tool previously computed all *site-specific* correction factors and supplementary characterisation quantities before sensor deployment.

Core physics outputs. The three multiplicative correction factors — κ_{topo} (topographic, 3-D ray-casting), κ_{muon} (muon field-of-view, analytical integral), and κ_{lulc} ($W(r)$-weighted WorldCover + OSM) — are combined into $\kappa_{\text{tot}} = \kappa_{\text{topo}}\kappa_{\text{muon}}\kappa_{\text{lulc}}$. Together with the lattice water l_w derived from SoilGrids clay fractions, they determine the reference count N_0 through the Desilets (2010) formula, which preserves full analytical invertibility and remains the primary calibration reference.

Supplementary characterisation. The seasonal climatology of atmospheric water vapour (f_{WV}), snow-water equivalent (SWE), and above-ground biomass hydrogen (AGBH) quantifies the operational corrections that will be required throughout the measurement period. The Macrostrat geological characterisation flags unusual lithologies that may bias the clay-based l_w estimate. The Topographic Wetness Index provides a spatially distributed map of expected moisture variability within the footprint, relevant to calibration planning.

Sampling design. The $W(r)$-optimised 26-point sampling plan, and the accompanying $r_{86}(\theta_v)$ dynamic-range plot, translate the footprint physics directly into field-campaign instructions. The plan minimises the variance of the footprint-averaged soil moisture estimate for a fixed number of samples.

Uncertainty. A combined uncertainty budget for the Italian Alps example yields $\sigma_{N_0}/N_0 \approx 5\%$, with the LULC classification uncertainty ($\sim 4\%$) and the soil lattice water ($\sim 3\%$) as the dominant terms. This demonstrates that accurate site characterisation can substantially reduce the systematic error in soil-moisture retrievals, and identifies where additional field effort (in-situ clay measurements, vegetation water content surveys) is most rewarding.

Outlook. Future extensions could include:

- Integration of LiDAR DEMs where available, to resolve sub-30-m terrain features relevant for κ_{topo}.
- A sensitivity analysis module that propagates input uncertainties through the full correction chain and reports quantitative bounds on N_0.
- Time-varying κ_{lulc} driven by satellite-derived LAI and canopy water content, enabling quasi-operational correction.
- Support for portable CRNS rovers [Schrön et al., 2017], where the footprint geometry changes with vehicle speed and the sampling plan module must be adapted accordingly.

The pipeline source code is openly available at `https://github.com/barbierimauro/site_characterization` under an open-source licence. All datasets used are publicly accessible without registration. We invite contributions and reports of site-specific validation datasets to expand the empirical basis for f_H assignments and l_w estimates.

References

R. Baatz, H. R. Bogena, H.-J. Hendricks Franssen, J. A. Huisman, C. Montzka, and H. Vereecken. An empirical vegetation correction for soil water content quantification using cosmic ray probes. *Water Resources Research*, 51(4):2030–2046, 2015. doi: 10.1002/2014WR016443.

K. J. Beven and M. J. Kirkby. A physically based, variable contributing area model of basin hydrology. *Hydrological Sciences Bulletin*, 24(1):43–69, 1979. doi: 10.1080/02626667909491834.

H. R. Bogena, J. A. Huisman, R. Baatz, H.-J. Hendricks Franssen, and H. Vereecken. Accuracy of the cosmic-ray soil water content probe in humid forest ecosystems: The worst case scenario. *Water Resources Research*, 49(9):5778–5791, 2013. doi: 10.1002/wrcr.20463.

H. R. Bogena, F. Herrmann, J. Jakobi, C. Brogi, H. Ilyas, J. A. Huisman, H. Vereecken, and A. Wahbi. Monitoring of snowpack dynamics with cosmic-ray neutron probes: a comparison of four approaches. *Frontiers in Water*, 2, 2020. doi: 10.3389/frwa.2020.00019.

D. Desilets, M. Zreda, and T. P. A. Ferré. Nature's neutron probe: Land-surface hydrology at an elusive scale with cosmic rays. *Water Resources Research*, 46(11):W11505, 2010. doi: 10.1029/2009WR008726.

European Space Agency and Airbus. Copernicus DEM – global and European digital elevation model. *ESA Open Access Hub*, 2022. doi: 10.5270/ESA-c5d3d65.

T. E. Franz, M. Zreda, T. P. A. Ferre, R. Rosolem, C. Zweck, S. Stillman, X. Zeng, and W. J. Shuttleworth. Measurement depth of the cosmic ray soil moisture probe affected by hydrogen from various sources. *Water Resources Research*, 48(8):W08515, 2012. doi: 10.1029/2012WR011871.

A. Hawdon, D. McJannet, and J. Wallace. Calibration and correction procedures for cosmic-ray neutron soil moisture probes located across Australia. *Water Resources Research*, 50 (6):5029–5043, 2014. doi: 10.1002/2013WR015138.

H. Hersbach, B. Bell, P. Berrisford, and et al. The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society*, 146(730):1999–2049, 2020. doi: 10.1002/qj.3803.

J. Iwema, R. Rosolem, M. A. Rahman, E. Blyth, and T. Wagener. Land surface model performance using cosmic-ray and point-scale soil moisture measurements for calibration. *Hydrology and Earth System Sciences*, 21(6):2843–2861, 2017. doi: 10.5194/hess-21-2843-2017.

M. Köhli, M. Schrön, M. Zreda, U. Schmidt, P. Dietrich, and S. Zacharias. Footprint characteristics revised for field-scale soil moisture monitoring with cosmic-ray neutrons. *Water Resources Research*, 51(7):5772–5790, 2015. doi: 10.1002/2015WR017169.

M. Köhli, M. Schrön, and U. Schmidt. Response functions for detectors in cosmic ray neutron sensing. *Nuclear Instruments and Methods in Physics Research A*, 902:184–189, 2021. doi: 10.1016/j.nima.2018.06.052.

J.-F. Pekel, A. Cottam, N. Gorelick, and A. S. Belward. High-resolution mapping of global surface water and its long-term changes. *Nature*, 540:418–422, 2016. doi: 10.1038/nature20584.

L. Poggio, L. M. de Sousa, N. H. Batjes, G. B. M. Heuvelink, B. Kempen, E. Ribeiro, and D. Rossiter. SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty, 2021.

R. Rosolem, W. J. Shuttleworth, M. Zreda, T. E. Franz, X. Zeng, and S. A. Kurc. The effect of atmospheric water vapor on neutron count in the cosmic-ray soil moisture observing system. *Vadose Zone Journal*, 12(3), 2013. doi: 10.2136/vzj2012.0120.

K. E. Saxton and W. J. Rawls. Soil water characteristic estimates by texture and organic matter for hydrologic solutions. *Soil Science Society of America Journal*, 70(5):1569–1578, 2006. doi: 10.2136/sssaj2005.0117.

M. Schrön, M. Köhli, L. Scheiffele, J. Iwema, H. R. Bogena, L. Lv, E. Martini, G. Baroni, R. Rosolem, J. Weimar, and et al. Improving calibration and validation of cosmic-ray neutron sensors in the light of spatial sensitivity. *Hydrology and Earth System Sciences*, 21 (10):5009–5030, 2017. doi: 10.5194/hess-21-5009-2017.

D. F. Smart and M. A. Shea. Fifty years of progress in geomagnetic cutoff rigidity determinations. *Advances in Space Research*, 44(10):1107–1123, 2009. doi: 10.1016/j.asr.2009.07.006.

L. Wang and H. Liu. An efficient method for identifying and filling surface depressions in digital elevation models for hydrologic analysis and modelling. *International Journal of Geographical Information Science*, 20(2):193–213, 2006. doi: 10.1080/13658810500433453.

D. Zanaga, R. Van De Kerchove, D. Daems, W. De Keersmaecker, C. Brockmann, G. Kirches, J. Wevers, O. Cartus, M. Santoro, S. Fritz, M. Lesiv, M. Herold, N.-E. Tsendbazar, P. Xu, F. Ramoino, and O. Arino. ESA WorldCover 10 m 2021 v200, 2022.

M. Zreda, W. J. Shuttleworth, X. Zeng, C. Zweck, D. Desilets, T. Franz, and R. Rosolem. COSMOS: the COsmic-ray Soil Moisture Observing System. *Hydrology and Earth System Sciences*, 16(11):4079–4099, 2012. doi: 10.5194/hess-16-4079-2012.

A List of Variables and Constants

Table 4 lists all symbols used in this technical note with their definitions, units, and the section or equation where they are first introduced.

Table 4: Variables and constants used in this note.

Symbol	Unit	Definition	Reference
Footprint geometry			
r_{86}	m	Radius enclosing 86% of neutron signal	Eq. (1)
z_{86}	cm	Depth enclosing 86% of neutron signal	Eq. (2)
$W(r)$	–	Radial sensitivity weight at distance r	Eq. (3)
A_1	$\mathrm{m\,m^3\,m^{-3}}$	Footprint radius coefficient (= 29.13)	Eq. (1)
A_2	$\mathrm{m^3\,m^{-3}}$	Footprint offset (= 0.0578)	Eq. (1)
Soil and hydrogen properties			
θ_v	$\mathrm{m^3\,m^{-3}}$	Volumetric liquid soil water content	Sec. 2
l_w	$\mathrm{g\,g^{-1}}$	Clay-mineral lattice water content	Eq. (14)
ρ_b	$\mathrm{g\,cm^{-3}}$	Dry soil bulk density	Eq. (2)
f_{clay}	$\mathrm{g\,g^{-1}}$	Clay mass fraction of soil	Eq. (14)
θ_{SOC}	$\mathrm{m^3\,m^{-3}}$	Hydrogen equivalent of SOC	Eq. (23)
ω_{SOM}	$\mathrm{g\,g^{-1}}$	Soil organic matter mass fraction	Eq. (23)
Desilets calibration			
N	cph	Measured neutron count rate	Eq. (4)
N_0	cph	Reference count at zero liquid water	Eq. (6)
a_0	–	Desilets parameter (= 0.0808)	Eq. (4)
a_1	–	Desilets parameter (= 0.372)	Eq. (4)
a_2	$\mathrm{m^3\,m^{-3}}$	Desilets parameter (= 0.115)	Eq. (4)
Correction factors			
κ_{topo}	–	Topographic correction	Eq. (7)
κ_{muon}	–	Muon field-of-view correction	Eq. (9)
κ_{lulc}	–	Land-use/land-cover correction	Eq. (11)
κ_{tot}	–	Total site correction ($= \kappa_{\mathrm{topo}}\kappa_{\mathrm{muon}}\kappa_{\mathrm{lulc}}$)	Eq. (13)
$f_H^{(c)}$	–	Hydrogen equivalence factor for class c	Tab. 1
Ray-casting (topographic correction)			
θ	rad	Zenith angle of ray	Eq. (8)
ϕ	rad	Azimuth angle of ray	Eq. (8)
α	rad	Local surface slope angle	Eq. (8)
L_{air}	$\mathrm{g\,cm^{-2}}$	Air mass path length of ray	Eq. (8)
L_{soil}	$\mathrm{g\,cm^{-2}}$	Soil path length of ray	Eq. (8)
λ_{air}	$\mathrm{g\,cm^{-2}}$	Neutron attenuation length in air (= 130)	Eq. (8)
λ_{soil}	$\mathrm{g\,cm^{-2}}$	Neutron attenuation length in soil (= 162)	Eq. (8)
Muon correction			
θ_z	rad	Muon zenith angle	Eq. (9)
$\psi(\phi)$	rad	Horizon elevation angle at azimuth ϕ	Eq. (9)
$\theta_{z,\max}(\phi)$	rad	Max unblocked muon zenith at ϕ	Eq. (9)
Site flux and pressure			
$N_{\mathrm{neut,site}}$	cph	Expected neutron count at site	Eq. (12)
N_{sl}	cph	Sea-level detector reference count	Eq. (12)
P	hPa	Site pressure	Eq. (12)
P_0	hPa	Reference pressure (= 1013.25)	Eq. (1)
β_n	$\mathrm{hPa^{-1}}$	Barometric coefficient (= 0.0077)	Eq. (12)
R_c	GV	Geomagnetic rigidity cutoff	Eq. (12)
f_{R_c}	–	Rigidity correction factor	Eq. (12)
α_H	$\mathrm{GV^{-1}}$	Rigidity coefficient (= −0.075)	Eq. (12)
Supplementary corrections			
ρ_{WV}	$\mathrm{g\,m^{-3}}$	Atmospheric water vapour density	Eq. (16)
f_{WV}	–	Water-vapour neutron correction	Eq. (21)
α_{WV}	$\mathrm{m^3\,g^{-1}}$	WV correction coefficient (= 0.0054)	Eq. (21)
SWE	mm	Snow-water equivalent	Eq. (17)
AGBH	mm	Above-ground biomass hydrogen equivalent	Eq. (22)
Terrain indices			
TWI	–	Topographic Wetness Index	Eq. (19)
A_s	m	Specific upslope drainage area	Eq. (19)
β	rad	Local slope angle	Eq. (19)

B Pipeline Pseudocode

The following pseudocode summarises the full execution sequence of the `crns-sitechar` pipeline (cf. `main.py`). Each numbered step corresponds to a function call that may read from or write to the on-disk cache.

Listing 1: CRNS site characterization pipeline — top-level pseudocode (main.py).

```
# -------------------------------------------------------
# Input parameters (from config or CLI)
# -------------------------------------------------------
LAT, LON = 46.4, 11.3 # decimal degrees
THETA_V_INIT = 0.20 # m^3/m^3, design moisture
N_SL = 2800 # cph, detector reference
OUTPUT_DIR = "output/"

# -------------------------------------------------------
# Step 1-2 Terrain
# -------------------------------------------------------
dem, dx, dy = download_dem(LAT, LON, buffer=3*r86_approx())
slope, aspect, horizon = compute_terrain_derivatives(dem, dx, dy)
twi = compute_twi(dem, slope, cache_dir=OUTPUT_DIR) # cached

# -------------------------------------------------------
# Step 3 Preliminary footprint parameters
# -------------------------------------------------------
r86_prelim = r86(THETA_V_INIT, P=site_pressure(ELEV))
z86_prelim = z86(THETA_V_INIT, rho_b=1.2, lw=0.0)

# -------------------------------------------------------
# Step 4 Land-cover and OSM
# -------------------------------------------------------
worldcover = download_worldcover(LAT, LON, r86_prelim)
osm = download_osm(LAT, LON, r86_prelim)

# -------------------------------------------------------
# Step 5 Soil properties (SoilGrids)
# -------------------------------------------------------
soil = download_soilgrids(LAT, LON)
rho_b = soil['bulk_density_gcc']
clay_gg = soil['clay_gg']
soc_gkg = soil['soc_gkg']
lw = 0.097 * clay_gg + 0.033 # Kohli 2021

# -------------------------------------------------------
# Step 6 Updated footprint with actual lw and rho_b
# -------------------------------------------------------
z86_act = z86(THETA_V_INIT, rho_b, lw)
r86_act = r86(THETA_V_INIT, P=site_pressure(ELEV))

# -------------------------------------------------------
# Step 7 kappa_topo (3-D ray-casting)
# -------------------------------------------------------
kappa_topo = compute_kappa_topo(dem, z86_act, rho_b,
                              n_theta=18, n_phi=72)

# -------------------------------------------------------
# Step 8 kappa_muon (horizon integral)
# -------------------------------------------------------
```

```
kappa_muon = compute_kappa_muon(horizon)

# -------------------------------------------------------
# Step 9 kappa_lulc (W(r)-weighted f_H)
# -------------------------------------------------------
lulc_result = get_lulc(worldcover, osm, dem, dx, dy,
                    r86_act, theta_v_init=THETA_V_INIT)
kappa_lulc = lulc_result['wc_kappa']

# -------------------------------------------------------
# Step 10 Site fluxes and N0
# -------------------------------------------------------
kappa_tot = kappa_topo * kappa_muon * kappa_lulc
flux = compute_site_fluxes(LAT, LON, ELEV, N_SL,
                       lw=lw, soc_gkg=soc_gkg,
                       rho_b=rho_b,
                       kappa_tot=kappa_tot)
N0 = flux['N0']

# -------------------------------------------------------
# Step 11 Atmospheric / snow seasonality (ERA5)
# -------------------------------------------------------
crns_corr = get_crns_corrections(LAT, LON,
                             lai=vegetation['lai_mean'])

# -------------------------------------------------------
# Step 12 Geological characterisation (Macrostrat)
# -------------------------------------------------------
geology = get_geology(LAT, LON)

# -------------------------------------------------------
# Step 13 ERA5-Land hourly soil moisture (era5sm.py)
# Downloads year-by-year from start_year to present.
# Results cached as hourly_YYYY.npz + monthly_agg.npz.
# -------------------------------------------------------
era5_sm = get_era5_soil_moisture(
    LAT, LON,
    cache_dir = OUTPUT_DIR,
    start_year = 2015,
    verbose = True,
)
# era5_sm keys: sm0_7_monthly_mean/std/min/max/nobs (12,)
# sm7_28_*, sm28_100_*
# sm_profile_mean (3,), sm_profile_depths (3,)
# sm0_7_current, years_available, site_dir

# -------------------------------------------------------
# Step 14 Downscaled soil moisture estimate (smphysics.py)
# Data fusion: ERA5 + SoilGrids + TWI + PISR + LULC.
# -------------------------------------------------------
sm_fused = fuse_soil_moisture(
    era5_res = era5_sm,
    soil_res = soil,
    twi_res = twi,
    climate_res = crns_corr,
    lulc_res = lulc_result,
)
# sm_fused keys: theta_monthly (12,), theta_monthly_low/high (12,)
```

```
# theta_annual_mean, theta_current
# FC_local, WP_local, SAT_local, sigma_total
# era5_raw_monthly, corrections

# -------------------------------------------------------
# Step 15 Sampling plan
# -------------------------------------------------------
sampling = compute_sampling_plan(r86_act, P=site_pressure(ELEV))
plot_sampling_plan(sampling, dem, dx, dy, dist_grid)

# -------------------------------------------------------
# Step 16 Assemble and write report
# -------------------------------------------------------
results = {
    'kappa_topo': kappa_topo,
    'kappa_muon': kappa_muon,
    'kappa_lulc': kappa_lulc,
    'kappa_tot': kappa_tot,
    'N0': N0,
    'lw': lw,
    'soc_gkg': soc_gkg,
    'crns_corrections': crns_corr,
    'geology': geology,
    'era5_sm': era5_sm,
    'sm_fused': sm_fused,
    'sampling': sampling,
    ...
}
write_report(results, OUTPUT_DIR)
save_json(results, OUTPUT_DIR / "results.json")
```

Figure 5 illustrates the data flow between modules.

Figure 5: Data-flow diagram of the CRNS site characterization pipeline. Rectangles denote processing steps; cylinders denote on-disk caches; rounded rectangles denote external data sources. Arrows indicate data dependencies.

Cache strategy. Each expensive step writes its result to a uniquely named file in `OUTPUT_DIR` (e.g. `dem_mosaic.npz`, `soilgrids_0_30cm.json`, `twi_cache_<hash>.npz`). At the start of each step the pipeline checks whether the output file already exists and, if so, loads it directly. This means that a run interrupted halfway can be resumed from the last successful step with no data re-download.

Configuration. All tunable parameters (site coordinates, N_{sl}, θ_v^{ref}, ray-casting resolution, ring fractions, download time-out) are collected in a single block at the top of `main.py` and are documented in Table 4. No external configuration file is required for standard use.